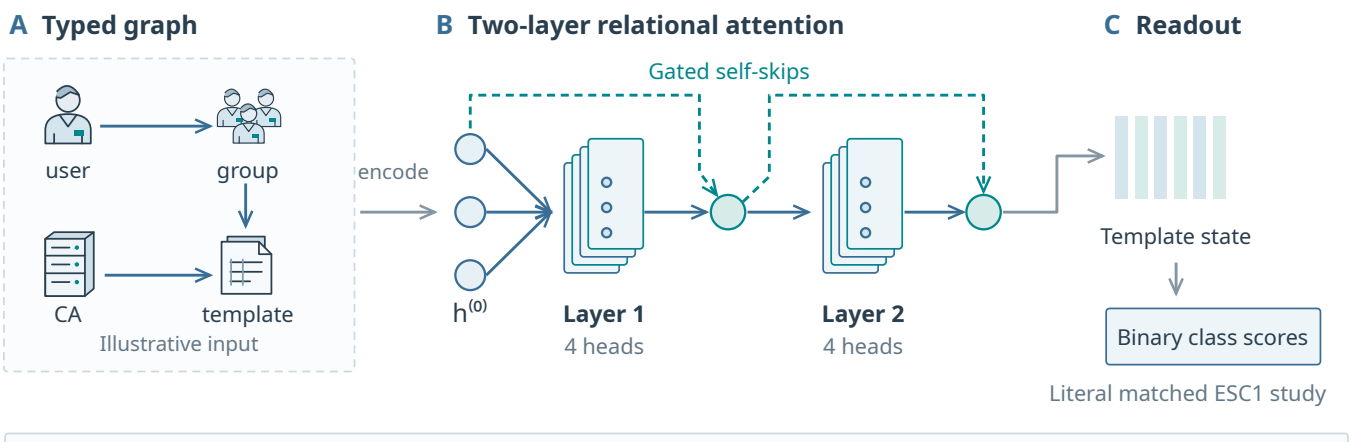
A Typed graph
user
group
CA
template
Illustrative input
encode
B Two-layer relational attention
Gated self-skips
h(0)
Layer 1
4 heads
Layer 2
4 heads
C Readout
Template state
Binary class scores
Literal matched ESC1 study
D Bilinear message mechanism
Source hu
Target hv
Relation map Wr
guv = sigmoid
(huT Wg hv + b)
Attention-weighted aggregation
plus gated self-skip
Authorization is checked separately.